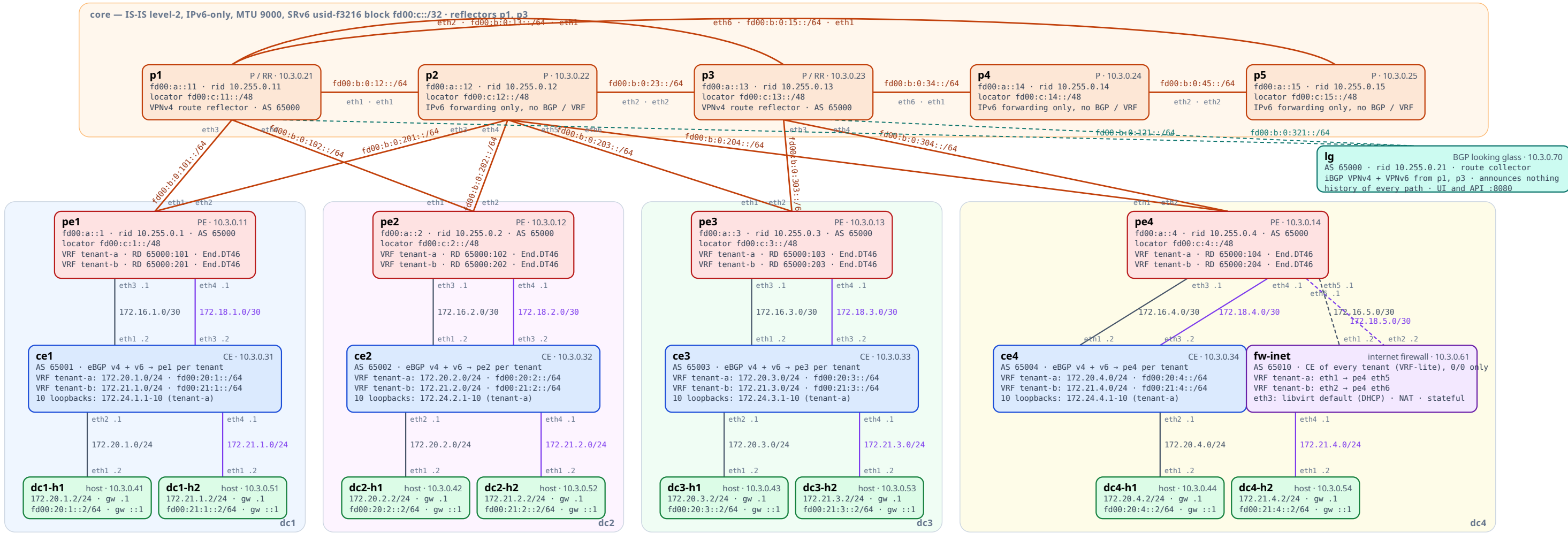


SRv6 WAN core lab — topology

Four VyOS PEs (one per data centre) dual-homed to a five-router VyOS P core (a triangle and an outer path p1-p5-p4-p3); IS-IS level-2 IPv6-only underlay carrying the SRv6 locators; BGP VPNv4 over SRv6 (End.DT4) reflected by p1; a VyOS CE per data centre with one VRF per tenant (tenant-a, tenant-b), each tenant on its own attachment circuit into its own VRF on the PE and with its own host per site. Tenants never meet. 23 VMs on one libvirt/KVM host. — core link (IPv6 /64, IS-IS, SRv6) — tenant-a access (IPv4) — tenant-b access (IPv4)



Addressing

Node	Role	OOB	Loopback	Router-id	IS-IS NET	SRv6 locator	AS
pe1	pe	10.3.0.11	fd00:a::1	10.255.0.1	49.0001.0000.0000.0001.00	fd00:c:1::/48	65000
pe2	pe	10.3.0.12	fd00:a::2	10.255.0.2	49.0001.0000.0000.0002.00	fd00:c:2::/48	65000
pe3	pe	10.3.0.13	fd00:a::3	10.255.0.3	49.0001.0000.0000.0003.00	fd00:c:3::/48	65000
pe4	pe	10.3.0.14	fd00:a::4	10.255.0.4	49.0001.0000.0000.0004.00	fd00:c:4::/48	65000
p1	p (RR)	10.3.0.21	fd00:a::11	10.255.0.11	49.0001.0000.0000.0011.00	fd00:c:11::/48	65000
p2	p	10.3.0.22	fd00:a::12	10.255.0.12	49.0001.0000.0000.0012.00	fd00:c:12::/48	–
p3	p (RR)	10.3.0.23	fd00:a::13	10.255.0.13	49.0001.0000.0000.0013.00	fd00:c:13::/48	65000
p4	p	10.3.0.24	fd00:a::14	10.255.0.14	49.0001.0000.0000.0014.00	fd00:c:14::/48	–
p5	p	10.3.0.25	fd00:a::15	10.255.0.15	49.0001.0000.0000.0015.00	fd00:c:15::/48	–
ce1	ce	10.3.0.31	–	172.20.1.1	–	–	65001
ce2	ce	10.3.0.32	–	172.20.2.1	–	–	65002
ce3	ce	10.3.0.33	–	172.20.3.1	–	–	65003
ce4	ce	10.3.0.34	–	172.20.4.1	–	–	65004
fw-inet	fw	10.3.0.61	–	10.3.0.61	–	–	65010
dc1-h1	host	10.3.0.41	–	–	–	–	–
dc2-h1	host	10.3.0.42	–	–	–	–	–
dc3-h1	host	10.3.0.43	–	–	–	–	–
dc4-h1	host	10.3.0.44	–	–	–	–	–
dc1-h2	host	10.3.0.51	–	–	–	–	–
dc2-h2	host	10.3.0.52	–	–	–	–	–
dc3-h2	host	10.3.0.53	–	–	–	–	–
dc4-h2	host	10.3.0.54	–	–	–	–	–
lg	lg	10.3.0.70	–	10.255.0.21	–	–	65000

Links: core fd00:b:0:<ab>::/64 (first end ::1), PE–CE 172.16.n.0/30 (PE .1), CE–host 172.20.n.0/24 (CE .1 = gateway). OOB network srv6-oob 10.3.0.0/24 (host 10.3.0.1), consoles 127.0.0.1:5301–5319. VRF tenant-a: table 100, RT 65000:100, RD 65000:100+pe# · VRF tenant-b: table 200, RT 65000:200, RD 65000:200+pe#. Tenant-b uses PE–CE 172.18.n.0/30 and LANs 172.21.n.0/24. Every tenant link is dual-stack: its IPv6 twin is fd00:X:Y::/64 of 172.X.Y.0 (fd00:16:1::/64, fd00:20:1::/64 ...), one End.DT46 per VRF serves both families.

Generated from lab.conf by docs/topology.py · <https://github.com/dcantor/srv6-core>

Packet walk: dc1-h1 → dc3-h1 (tenant-a, dc1 → dc3)

- dc1-h1** 172.20.1.2 sends to 172.20.3.2 via its gateway **ce1** (172.20.1.1).
 - ce1** has 172.20.3.0/24 from pe1 in its VRF tenant-a over that VRF's eBGP session → forwards to **pe1** 172.16.1.1 (VRF tenant-a on the PE too).
 - pe1**: VRF route 172.20.3.0/24 = encap seg6 segs 1 [fd00:c:3:e0XX::] — the End.DT4 SID pe3 exported with the VPNv4 route (RD 65000:103, RT 65000:100, next hop fd00:a::3) via the route reflectors p1 and p3 (the PE keeps both copies; losing one reflector changes nothing). Outer IPv6 fd00:a::1 → fd00:c:3:0:X:: + SRH.
 - p2** (the only shortest path west → east) forwards plain IPv6 towards pe3's locator fd00:c:3::/48 learned from IS-IS — no VRF, no IPv4 knowledge.
 - pe3**: local SID seg6local End.DT4 vrf table tenant-a decapsulates and looks the inner packet up in the VRF → **ce3** 172.16.3.2 → **dc3-h1**.
 - A packet from **dc1-h2** (tenant-b) takes the same core path but enters through CE VRF tenant-b, the second attachment circuit, PE VRF tenant-b and pe3's *other* End.DT4 SID; it can never reach a tenant-a address because no tenant-a route exists in any tenant-b table (different RTs).
 - Reply mirrors the path with pe1's SID. Forwarded packets need the locator block leaked into the VRF table (static route6 fd00:c::/40 ... vrf default) because Linux scopes the encapsulation's outer lookup to the ingress VRF.

Local SIDs on a PE (pe1)

SID	Behaviour	Installed by
fd00:c:1::/48	uN (End, NEXT-C-SID flavour: shift 16 bits, forward)	IS-IS
fd00:c:1:e000::, fd00:c:1:e001::	uA — End.X per core adjacency	IS-IS
fd00:c:1:e002::, fd00:c:1:e003::	uDT46 — End.DT46 → VRF tenant-a / tenant-b (one per tenant, IPv4 and IPv6)	BGP (sid vpn export auto in each VRF)

uSID (usid-f3216): block 32 · node 16 · function 16 bits, /48 locators from fd00:c::/32; a steered path p1 → p3 → pe3 is the single segment fd00:c:11:13:3:e001::. Function values are allocated by FRR at run time.